

Product Brief: Helios Sensor Array

Key Specifications

Sampling rate: **4 kHz** per channel, 32 channels total.

Noise floor: 0.8 μV^2 under shielded conditions.



Image 1: Thermal distribution across the array face.

Deployment Notes

The array ships pre-calibrated. Field recalibration is only required after mechanical shock or when operating temperature moves outside the rated band for more than an hour. Mount the unit on the vibration isolators provided; rigid mounting couples structural noise directly into the low-frequency channels and can mask weak signals entirely.

Integration Checklist

Verify supply ripple stays below the published limit before connecting the analog front end. Route the sensor harness away from switching converters. Confirm that the acquisition host applies the vendor timestamp rather than re-stamping frames on arrival, since downstream alignment assumes source timestamps.

After first power-up, run the built-in self test and archive the report it produces. The report is the baseline used by support when diagnosing drift claims during the warranty period.

² Measured with inputs shorted at the connector.